Research conditions and distribution

Experimental group: IBL + VL. Comparison group: no prelesson lecture and no VL operation; students self-study using A/B worksheets. Paper follows explanation → worked examples → guided practice → independent application → checking/correction, with shared content and reasonable science-process practice rather than the VL's delayed concept reveal.

Stage	15 min	20 min	Distribution and support
A: concepts and images	3	4	Give A at the start; no prelesson lecture.
A: worked examples	2	2	Students read independently.
B1–B4: guided practice	5	7	Give B; use the shared printed hints.
B5: independent application	2	3	A remains available; no individual answers.
C: check and correct	3	4	Give C only after B is finished and first answers retained.

Choose one common 15–20-minute total for both groups, including reading, answering and feedback. These are two planning schedules, not measured durations. Record actual time and deviations; paper actions or completion are not VL abilities.

Teacher clarification and feedback

During A/B, clarify only procedures, vocabulary and diagram-reading conventions using the same planned hints. Do not add a lecture or give individual answers or growth directions. Record extra conceptual teaching as a protocol deviation. Retain completed B, then give C and allow content feedback based on C; record corrections separately. D and the full pack are for teachers only.

Shared goals, separate scoring and outcomes

Share the context, three investigations, model images, core concepts and goals in observing, variables, fair comparisons, measurement, inference and communication; teaching sequence differs. Paper does not score advance guesses, clicks, demonstrations, animation completion or virtual dragging. D's 14 points are formative and must not be compared directly with VL's 32. Keep common SPSAI/CKT pre/post-tests separate and unchanged; do not add experimental-only opinion or interview items.

VL1 informs PDF interface and layout only. Use displayed C for below-tip shading and D for cutting; keep legacy C/D records and compatibility unchanged. Generation, preview and download do not create learner records or write answers, cloud data, events or teacher Excel.

Student checking answers

C1/C2 provide all closed-question answers and key points for open responses in B; the complete pack includes both C pages. For teacher judgement, accept scientifically equivalent wording rather than exact copying.

Observation, labels, variables and controls

P1: straighter initially, then bends towards the light on the right. P2: A uncovered, B tip shaded, C shaded below the tip. P3: shading is IV; bending is DV; light direction/intensity, species/initial size/condition, time, temperature/water are CVs. The first three assumptions apply; same species does not remove size control.

A is uncovered and provides a baseline without shading in the same environment. A–B changes tip shading; A–C changes shading below the tip. Keep conditions such as initial size and temperature the same to avoid other influences.

Data and conclusions from all three investigations

Main comparison: bending direction

A	No light blocked	Bends left
B	Light blocked at the tip	No clear bending
C	Light blocked below the tip	Bends left

Extension one: presence or absence of elongation（延長）

A	No light blocked	Yes
D	Tip removed	No

Extension two: angle and direction

E	Blank agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
F	Treated agar（瓊脂） • Centre	0°（Teaching model）	No clear bending
G	Treated agar（瓊脂） • Left	35°（Teaching model; ±2°）	Bends right
H	Treated agar（瓊脂） • Right	35°（Teaching model; ±2°）	Bends left

3. Which part may sense light from one side? Tip

1. Compare A and B: What changes when light is blocked at the tip? No clear bending towards the light.

2. Compare A and C: Does it still bend when the lower part receives no direct light? It can still bend towards light.

1. What does comparing elongation（延長） in A and D support? The presence of the tip also affects the coleoptile（胚芽鞘）’s growth / elongation（延長）.

1. What does the comparison of E and F support? The tip may produce a growth-promoting effect that can be transferred.

2. What does the comparison of F, G and H support? The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives more substance X, causing cells on the left to **elongate（延長）** more and the coleoptile（胚芽鞘） to bend to the right side.

A–D: whether the tip is present. E–F: whether agar（瓊脂） previously touched the tip, with both in the centre. F–G–H: treated agar（瓊脂） position, with the same agar（瓊脂） type. The main comparison changes shading treatment.

Reflection and correction

Right-side light makes the left the shaded side. The tip senses light; more auxin（生長素） on the left promotes greater left-cell elongation（延長）, so the coleoptile（胚芽鞘） bends towards the light on the right.

C2 correction: retain first answers, identify a correction and one data-based reason. If initially correct, provide supporting evidence. The causal chain must show greater elongation（延長） on the shaded side, not the lit side.

Cutting creates a wound and removes tissue, so it cannot establish a unique mechanism alone. Dark agar（瓊脂） results support a transferable growth-promoting effect and unequal elongation（延長）. Bending is not caused by light during incubation; the results neither measure auxin（生長素） concentration directly nor identify the substance.

VL-to-paper question mapping

Original / ID	Treatment	New question	Shared goal	Reason
P1 • Initial shape and direction observation	Retain / static	Describe changes in shape and direction in the window-side images.	Observation and recording	Same context; images replace animation.
E1 • Original main-inquiry hypothesis and reason	Worked example	Evaluate a given prediction: blocking light below the tip may still allow bending towards light.	Testable prediction and reasoning	Teach the mechanism first; model an evidence-based prediction without scoring an original hypothesis.
P2 • Autonomous setup drawing / written design	Rewrite / static	Label the shading treatments in three provided diagrams.	Identify apparatus and treatments	Provide the same setups for scientific labelling; do not equate this with virtual operation.
P3 • Main-inquiry variables and assumptions	Retain	Select main-inquiry variables and all applicable assumptions.	Classify and control variables	Keep variable choices and assumptions about initial size, temperature and unrestricted growth.
P4 • Autonomous control design	Equivalent rewrite	Explain the role of uncovered A and fair comparisons A–B / A–C.	Fair comparison and controls	Supply the control while retaining reasoning about baseline and controlled conditions.
P5 • Main A / B / C direction records	Static	Circle the bending direction after incubation for each setup.	Observation and recording	Shared before/after models and original direction options.
P6 • Compare A and B: what changes when the tip is shaded?	Retain	Compare A and B: what changes when the tip is shaded?	Comparison and inference	Keep the question and options; apply taught concepts to data.
P7 • Compare A and C: does it bend without direct light below the tip?	Retain	Compare A and C: does it bend without direct light below the tip?	Comparison and inference	Use displayed C without renumbering legacy D records.
P8 • Which part may sense light from one side?	Retain	Which part may sense light from one side?	Infer light sensing from evidence	Keep the A/B and A/C evidence chain.
E2 • Original extension-one prediction and reason	Application example	Use the tip's growth-promoting role to evaluate a cut-tip prediction and its limits.	Reasoned prediction	Teach the role and wound caveat rather than scoring an advance guess.

VL-to-paper question mapping

Original / ID	Treatment	New question	Shared goal	Reason
P9 • A/D elongation（延長） records and conclusion	Retain / static	Record whether A and D elongate（延長）, then conclude about the tip's role.	Qualitative observation and inference	Shared intact A / cut-tip D images and yes/no options.
E3 • Original extension-two prediction and reason	Explanation	Explain how agar（瓊脂） carries an effect and how placement relates to differential elongation（延長）.	Model understanding and prediction	Teach the method before independent interpretation of dark E–H.
P10 • E–H adjustable-pointer measurement and direction	Static	Read static protractors and record direction separately.	Measurement and recording	Retain the 0/0/35/35 model; image reading is not virtual-operation scoring.
P11 • What does comparing E and F support?	Retain	What does comparing E and F support?	Evidence for a transferable effect	Both are central; only prior tip contact differs.
P12 • What does comparing F, G and H support?	Retain	What does comparing F, G and H support?	Placement and bending	Same treated agar（瓊脂）, different positions; no light during dark incubation.
P13 • Extension-two cloze on differential elongation（延長）	Retain	Select more/less X on the left, more/less elongation（延長）, and left/right bending.	Reason about differential elongation（延長）	Keep options as circling choices rather than dropdowns.
P14 • Fair-comparison aims across three investigations	Equivalent application	Identify the independent variable separately for A–D, E–F and F–G–H.	Transfer variable classification	Distinguish shading, tip removal, agar（瓊脂） conditioning and placement.
P15 • Use the learning points to explain the right-bending coleoptile（胚芽鞘） in the diagram.	Retain / independent application	Use the learning points to explain the right-bending coleoptile（胚芽鞘） in the diagram.	Integrate concepts in an explanation	Link right-side light, shaded left, auxin（生長素） and differential elongation（延長）, not hypothesis revision.
C1 • Learning points revealed at the end	Move to A; C checks/corrects	Read concepts from the start; after B, check answers and record corrections separately.	Conceptual understanding and correction	Fits the self-study condition without a prelesson lecture.

Paper formative assessment (separate from VL)	Points	Paper evidence and criteria
Observation and recording	0–2	B2/B3/B4/B5: direction, presence/absence of A/D elongation（延長） and window-side shape. 2: accurate and complete; 1: partly accurate; 0: no valid record.
Variable classification	0–2	B1/B5: main shading IV/DV/CVs and independent variables for cutting, agar（瓊脂） treatment and placement. 2: correct across comparisons; 1: partly correct; 0: not identified.
Fair comparison and assumptions	0–2	B1: labels, A as baseline, initial size and temperature controls, unrestricted growth under shading. 2: sound and complete; 1: partial; 0: no fair-comparison understanding.
Evidence-based inference	0–2	B2/B3: conclusions from A/B, A/C, A/D, E/F and F/G/H. 2: distinguishes the effects each comparison supports; 1: partial; 0: inconsistent with data.
Static protractor reading	0–2	B3/B4: E/F require 0°; G/H accept 33–37°, 0.5 per setup. Assess image reading only, not dragging or virtual-operation skill.
Scientific communication and correction	0–2	B1/C2: fair-comparison reasoning and evidence for separately recorded corrections. 2: clear reason–evidence link; 1: partial; 0: no valid link. Keep B first answers; assess C corrections separately.
Concept application	0–2	B4/B5: left agar（瓊脂） → more X → greater left elongation（延長） → right bend; right light → shaded left → more auxin（生長素） → greater left elongation（延長） → right bend. 2: both causal chains correct; 1: partial; 0: reversed or unexplained.
Total and use	14	For paper formative feedback only. Not the VL 32-point score, original-prediction or completion score, and not an outcome test. Evaluate research outcomes with the approved common SPSAI/CKT assessments.